

Core: Requirements

CHAPTER 2: REQUIREMENTS DEFINED

Chapter Objectives

In this chapter, we will develop an understanding of requirements including:

- Moving from deliverables to requirements
- The Requirements Gathering Cycle
- Defining the word “requirement”
- Categorizing requirements
- Identifying types of requirements

Chapter Concepts

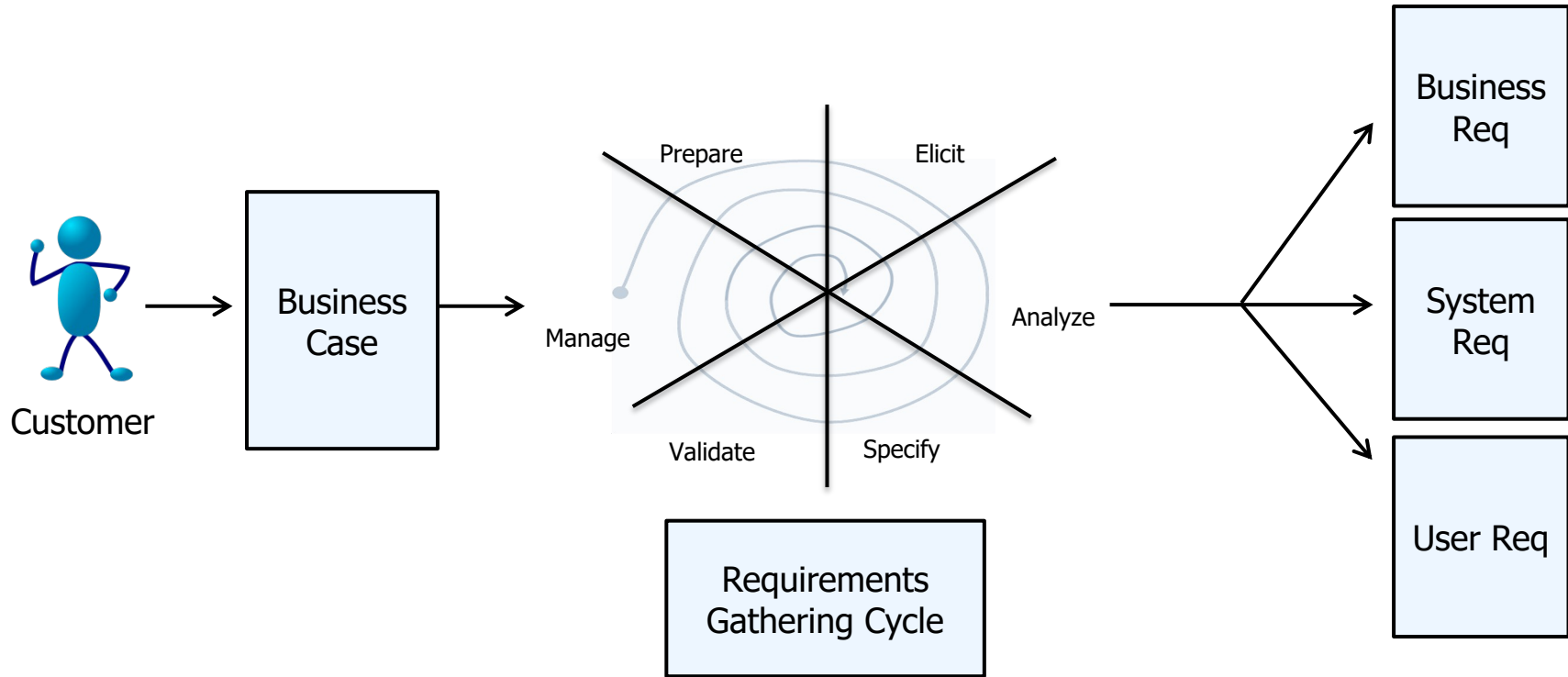
From Deliverables to Requirements

The Requirements Gathering Cycle

Requirements Defined

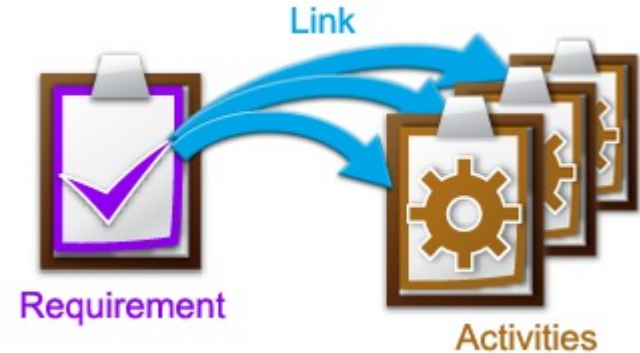
Types of Requirements

From Deliverables to Requirements

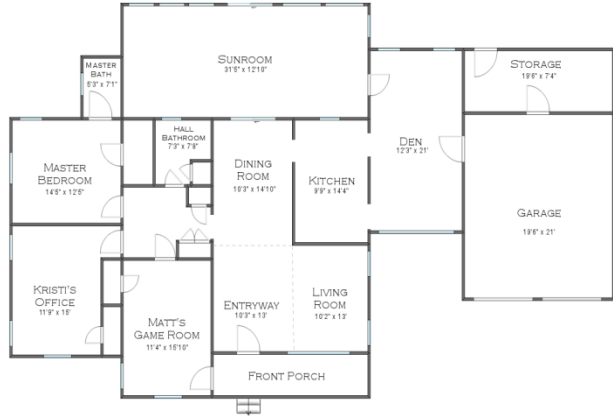


From Deliverables to Requirements (continued)

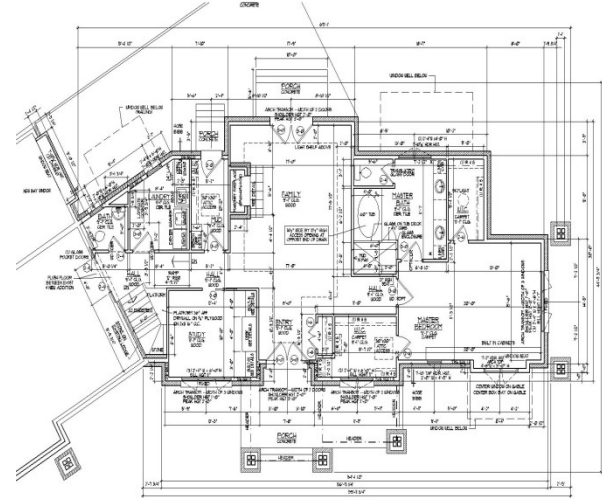
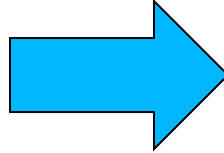
- Business case specifies contracted deliverables
 - Tangible (they are concrete and verifiable)
 - Metrics of quality (measures will be used to evaluate their quality)
- Analysts must now translate deliverables into formal requirements
 - Like an architect translating a floorplan into blueprints
- Requirements are then verified and validated using metrics and quality standards



Floor Plan to Blueprint



Floor Plan



Blueprint

Deliverables vs. Requirements

Deliverable	Requirement
Specified in Business Case	Specified in multiple requirement docs
Created by business partner	Created by Systems Analyst
High level, lacking in details	Highly detailed
Uses natural English	Uses structured syntax
Often less than five in total	Can be hundreds
Same approach in Waterfall or Agile	Very different approaches
Not prioritized, all required	Often prioritized because of cost and time
Only focused on business needs	Contains non-functional requirements as well

Chapter Concepts

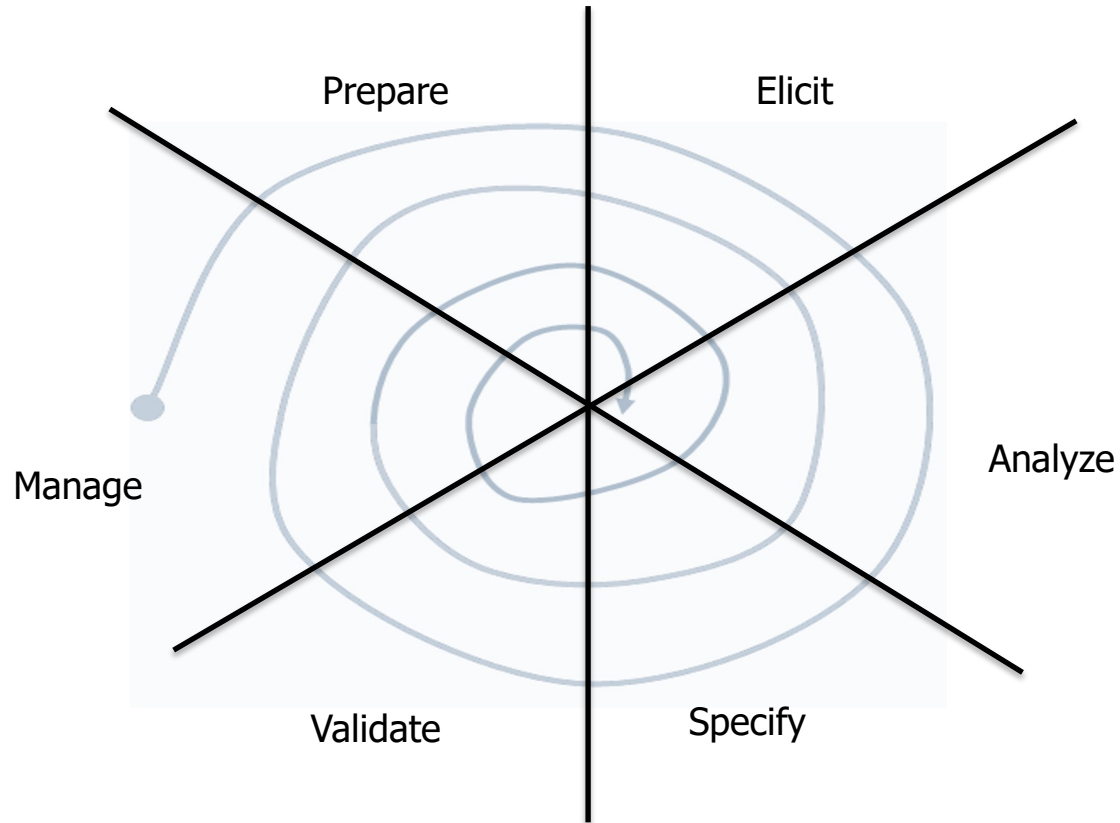
From Deliverables to Requirements

The Requirements Gathering Cycle

Requirements Defined

Types of Requirements

Requirements Gathering Cycle



Stages of the Gathering Cycle

- ◆ Prepare
 - Determine what needs to be gathered and how it will be done
 - Create question plans and sequence of interviews
- ◆ Elicit
 - Gather information via interview, survey, observation, etc.
 - Resolve requirement conflicts as needed
- ◆ Analyze
 - Move from stated requirements to real requirements
 - Categorize, prioritize, schedule (Agile), estimate
 - Create Use Cases, Process Maps, Flowcharts as needed



Stages of the Gathering Cycle (continued)

◆ Specify

- Write well-formed, well structured, good requirements
- Determine how to organize requirements (System, User, Business, etc.)
- Develop a numbering system to track requirements
- NB: Change/scope control must be available at this point

◆ Validate

- Verify and validate requirements with stakeholders
- Assure that traceability has been implemented
- Conduct inspections, reviews, or walkthroughs, as appropriate
- Begin planning test cases
- This is a key QAE role



◆ Manage

- Ongoing process work to handle changes, manage expectations, give status, etc.

Chapter Concepts

From Deliverables to Requirements

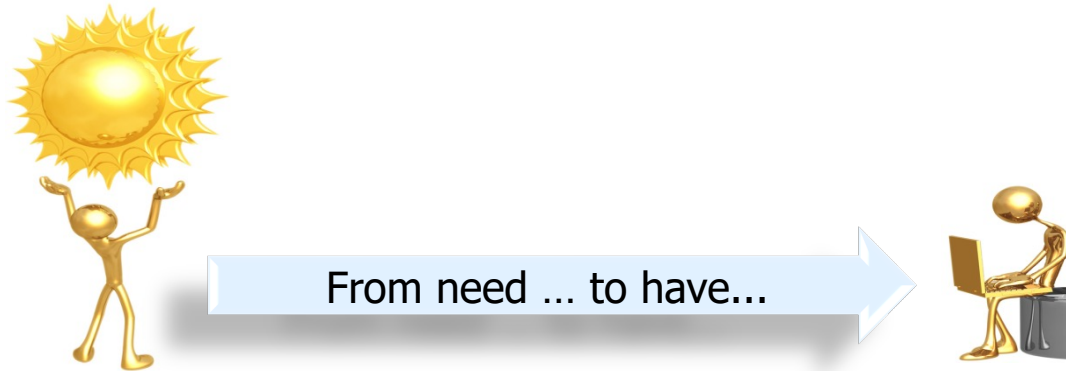
The Requirements Gathering Cycle

Requirements Defined

Types of Requirements

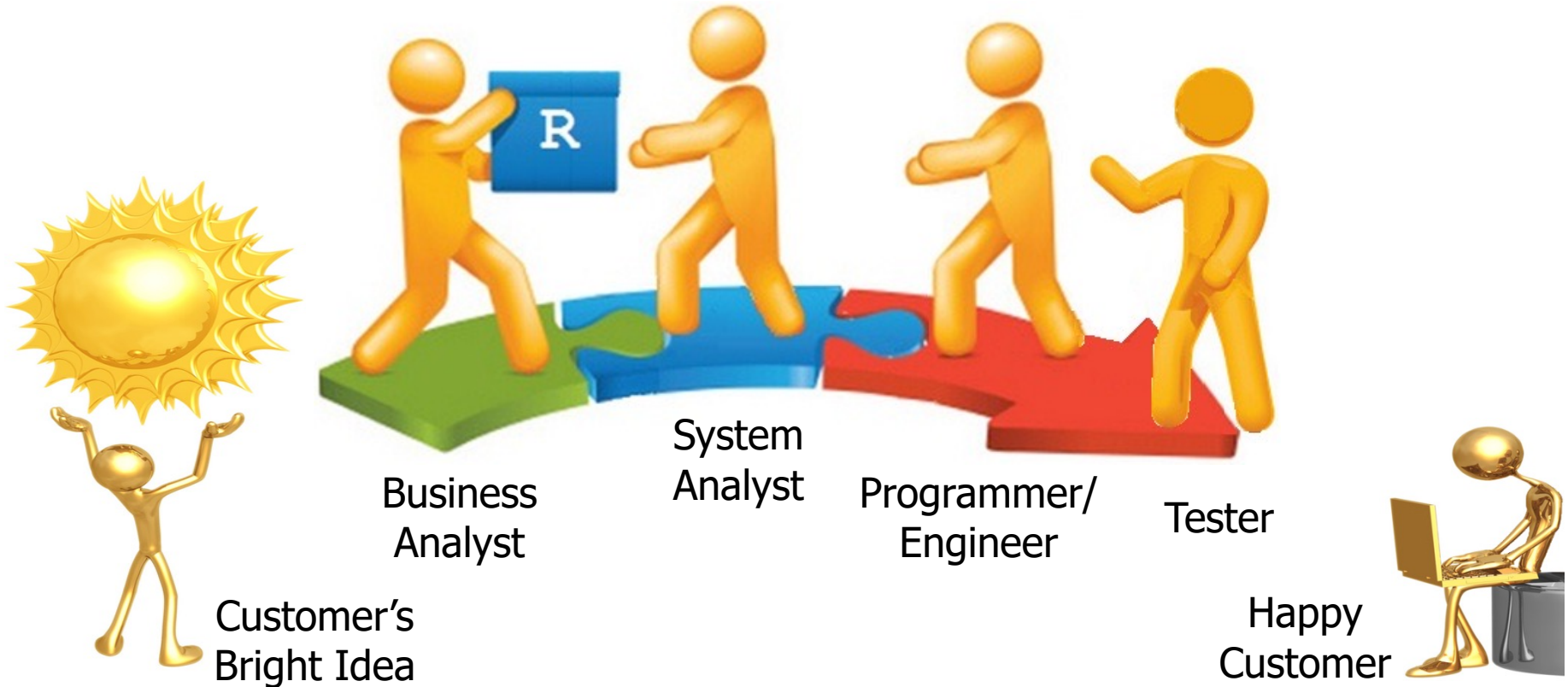
Definition of a Requirement

- A condition or capability
 - Needed by a stakeholder to solve a problem or achieve an objective
 - Must be met or possessed by a solution or solution component to satisfy a contract, standard, specification, or other formally imposed document
 - A documented representation of a condition or capability



Source: Business Analysis Body of Knowledge (BABOK®) version 2.0, www.iiba.org

Requirements Are About Communication



Software Requirements

- Software requirements define what the system should do and the constraints or expectations of the environment
 - Capabilities
 - Features
 - Functionality
- Identifies WHAT capabilities are needed by the system
 - Not HOW it will be implemented

The hardest single part of building a software system is deciding precisely what to build. No other part of the conceptual work is as difficult as establishing the detailed technical requirements, including all the interfaces to people, to machines, and to other software systems. No part of the work so cripples the resulting system if done wrong. No other part is more difficult to rectify later.

—Brooks, F.; *No Silver Bullet: Essence and Accidents of Software Engineering*, Computer 20, 4 (April 1987)

Chapter Concepts

From Deliverables to Requirements

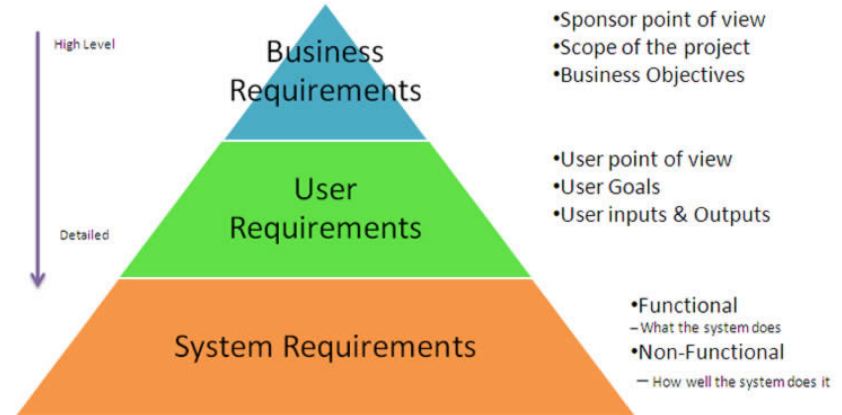
The Requirements Gathering Cycle

Requirements Defined

Types of Requirements

Requirements Target an Audience

- ◆ Requirements are written for a specific audience
 - Business users
 - Programmers
 - Data analysts
- ◆ Requirements are written in increasingly targeted levels of detail for a specific audience



Requirement Types Explained:

Business Requirements

◆ Business Requirements (BR)

- High-level objectives of the organization or customer requesting the system or product
- Normally tied directly to business case deliverables
- Objective is to be used to validate that the customer's needs have been properly, and fully, identified
 - ◆ Language favors ease of understanding over precision
 - ◆ Focus is on achieving business goals with respect to the business case
- Example of business requirements:
 - ◆ *Sales staff shall be notified of contracts expiring within the upcoming 90 days on a daily basis*
 - ◆ *The system shall automatically calculate sales taxes based on relevant sales tax laws*

Requirement Types Explained:

User Requirements

◆ User Requirements (UR)

- User goals or tasks that users must be able to perform with a system, or statements of the user's expectations of system quality
- Defines the user experience in terms of what the user wants the system to do
- Should relate directly to deliverables from the business case
- Should express the needs, not the solution!
- Normally written using a less formal structure and syntax
 - ◆ This makes it easier to reach agreement with users/stakeholders
 - ◆ Like Business Requirements, focus is on ease of understanding rather than precision

◆ Examples:

- *The User may order any product in the existing inventory*
- *The System shall display the Shopping Cart screen when the User clicks on the Checkout icon*
- *The User shall input their trading password before any buy/sell transaction can be processed*

Requirement Types Explained: System/Software Requirements

➤ System/Software Requirements (SR)

- Two possible definitions:
 1. The set of hardware and system software components needed to efficiently and effectively run a business/user process (i.e., software)
 2. A set of capabilities or services which a system must provide in order to satisfy the deliverables of a business case
- For the purposes of this program, we will use definition #2
- System/software requirements represent the “bridge” between BR/UR and the actual system design
- Normally written using formal syntax and structure
- Focus is on precision and accurately reflecting the behavior of the system
- Frequently differentiated between “functional” and “non-functional”

Requirement Types Explained: System/Software Requirements (continued)

➤ Examples of System/Software Requirements

- *The System shall validate the User's debit card using PIN validation techniques*
- *The System shall be capable of operating between -10C - +30C*
- *The System shall create an entry in an Exception Log after any User fails to login after three attempts*
- *The System shall provide weekly transaction reports for all clients for each Broker*
- *The System shall be available 24 x 7 with an availability not less than 99.97% on an annual basis*

Instructor-Led Activity 2.1: Requirements by Type

◆ Identify the following as **BR**, **UR**, or **SR**

#	Type	Requirement
1		Meridian Financial shall provide consumers with a do-it-yourself bond purchase system.
2		The bond search results page shall list results in high-relevance order appearing first.
3		A new bond system shall increase the number of bond transactions driving an increase in revenue.
4		Users shall access the bond system 7 days a week, 24 hours a day.
5		Bond owners shall be able to link their equity account credit line to their bond account credit line.
6		Customer information shall be shared between the equity system and the bond system.

Instructor-Led Activity 2.1: Requirements by Type (continued)

◆ Identify the following as **BR**, **UR**, or **SR**

#	Type	Requirement
1	BR	Meridian Financial shall provide consumers with a do-it-yourself bond purchase system.
2	SR	The bond search results page shall list results in high-relevance order appearing first.
3	BR	A new bond system shall increase the number of bond transactions driving an increase in revenue.
4	UR	Users shall access the bond system 7 days a week, 24 hours a day.
5	UR	Bond owners shall be able to link their equity account credit line to their bond account credit line.
6	SR	Customer information shall be shared between the equity system and the bond system.

Categories Within Requirement Types

- Assumptions (A)
- Business Rules (B)
- Constraints (C)
- Dependencies (D)
- Functional Requirements (FR)
- Non-Functional Requirements (NF)



Assumptions

- A thing that is accepted as true or as certain to happen, without proof
- Undocumented assumptions are the enemies of good analysis
 - Assumptions we make about the information (or gaps in information) that we have available to us
 - Assumptions stakeholders make about our knowledge of the process or understanding
- Assumptions are a normal part of a project
 - Must be documented
 - Must be monitored to ensure it remains true
 - All project assumptions need to be monitored as project risks

A14	Status = Confirmed	• This project assumes the BLB system will be free of charge to Meridian Financial customers.
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Business Rules

- ◆ A collection of the business logic in use by the application

B202	Password Criteria	• Password length: 5-32 • Password valid chars are: abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890!@#\$%^&*()_+~-\ {}[].,?/ • Password must contain at least two digits and one alpha
B203	Phone Numbers Criteria	- Have an area code - Be no longer than 10 digits - Must be in this format: XXX-XXX-XXXX - Must be valid
B204	Email Criteria	- Format of "<u>xxxxxxxxxx@xxxxx.com</u>" - Must be valid

Constraints

- ◆ Operating boundaries for the project
- ◆ Includes items like:
 - Budget
 - Resources
 - Timeline
 - Systems that must be used

C22	Budget	This project is capped at \$1,000,000
C23	Delivery Date	This project has a regulatory delivery date of 12/31/2027
C24	Systems	This project shall use existing infrastructure

Dependencies

- ◆ Things that the project's product and timeline require to be in place for the successful delivery of the product
- ◆ May include:
 - Vendor dependencies
 - Resource dependencies
 - Other project deliverable dependencies
- ◆ All project dependencies need to be monitored as project risks

D88	Vendor	• Vendor XYZ shall provide a card swipe hardware device capable of reading a magnetic stripe with 99% accuracy
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Functional Requirements

- ◆ The capabilities required by stakeholders
- ◆ May include:
 - Inputs
 - The behavior a stakeholder can perform or a system should provide
 - Outputs created
 - Services
 - Tasks
 - Functions

FR	Accounting	• The accounts payable users shall be able to generate a list of past-due invoices
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Non-Functional Requirements

- ◆ Requirements that specify the environmental characteristics in which the functional requirements will be operating
- ◆ Often called quality attributes or qualities of the system
- ◆ Includes:
 - Observable execution qualities such as availability and performance
 - Design evolution qualities such as maintainability, extensibility, and scalability

NFR	Accounting	• The invoicing system shall be available 23.5 hours each day
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Areas of Non-Functional Requirements

- ◆ Reliability
- ◆ Availability
- ◆ Maintainability
- ◆ Performance
- ◆ Accessibility
- ◆ Environmental conditions
- ◆ Ergonomic
- ◆ Safety
- ◆ Security
- ◆ Facility requirements
- ◆ Transportability
- ◆ Training
- ◆ Documentation
- ◆ External interfaces
- ◆ Testing
- ◆ Quality provisions
- ◆ Policy and regulatory
- ◆ Compatibility to existing systems
- ◆ Standards and technical policies
- ◆ Conversion
- ◆ Growth capacity
- ◆ Installation
- ◆ Migration
- ◆ Accountability
- ◆ Auditability
- ◆ Traceability
- ◆ Privacy
- ◆ Globalization
- ◆ Localization
- ◆ Scalability
- ◆ Portability
- ◆ Backup, restore, and recovery
- ◆ Compliance
- ◆ Efficiency
- ◆ Modifiability
- ◆ Usability

Instructor-Led Activity 2.2: Requirements by Category

➤ Identify the following as A, B, C, D, FR, or NF?

#	Category	Requirement
1		Meridian Financial shall provide consumers with a do-it-yourself bond purchase system.
2		The bond search results page shall list results in high-relevance order appearing first.
3		A new bond system shall increase the number of bond transactions driving an increase in revenue.
4		Users shall access the bond system 7 days a week, 24 hours a day.
5		Bond owners shall be able to link their equity account credit line to their bond account credit line.
6		Customer information shall be shared between the equity system and the bond system.

Instructor-Led Activity 2.2: Requirements by Category (continued)

➤ Identify the following as A, B, C, D, FR, or NF?

#	Category	Requirement
1	FR	Meridian Financial shall provide consumers with a do-it-yourself bond purchase system.
2	FR	The bond search results page shall list results in high-relevance order appearing first.
3	A	A new bond system shall increase the number of bond transactions driving an increase in revenue.
4	NF	Users shall access the bond system 7 days a week, 24 hours a day.
5	FR	Bond owners shall be able to link their equity account credit line to their bond account credit line.
6	C, D	Customer information shall be shared between the equity system and the bond system.

Stated Requirements vs. Real Requirements

- ◆ Frequently what the Analyst is told is not the real requirement
 - Some stakeholders demand a particular technology
 - Some stakeholders don't understand their problem
 - Some stakeholders have hidden agendas
 - Etc.
- ◆ The Team must look past the stated requirement
 - What is the real problem under investigation?
 - In what way will the stated requirements address the problem?
 - Has 3P analysis been performed for this stakeholder?



Activity 2.3: Categories of Requirements



Goal	Identify categories of requirements
Team Format	Three teams
Materials	The next slide
Tasks	1. Identify each of the listed requirements as one of: • A – an assumption • B – a business rule • C – a constraint • D – a dependency • FR – a functional requirement (FR) • NF – a non-functional requirement (NF)
Deliverable	Requirements identified by category
Duration	20 minutes

Activity 2.3: Categories of Requirements (continued)



#	Category	Requirement
1		ABC Airlines shall transfer time-consuming repetitive tasks from airline staff to airline travelers.
2		Users shall be able to check in at any partner airline kiosk.
3		Airline shall provide automated check-in services at airports for travelers with existing airline reservations.
4		Airport kiosks shall not support a Braille interface.
5		Check-in kiosks shall be available at airports 24x7.
6		The System shall assign the traveler 100 FrequentTraveler credits when check-in is performed at an airport kiosk.
7		The System shall validate the FrequentTravelerID according to the following criteria: 10 characters, can be blank, alphanumeric, starts with a letter, upper-case letters, will not include the letter O.
8		The System shall use the traveler's FrequentTravelerID as the primary key for all kiosk-initiated transactions.
9		Traveler may check-in at a kiosk when flight departure time is greater than 60 minutes and less than 24 hours.
10		Traveler check-in shall be available at all major airports by the end of next fiscal year.

Activity 2.3: Categories of Requirements (continued)



#	Category	Requirement
1	BR, FR, A	ABC Airlines shall transfer time-consuming repetitive tasks from airline staff to airline travelers.
2	UR, FR, A	Users shall be able to check in at any partner airline kiosk.
3	BR, FR	Airline shall provide automated check-in services at airports for travelers with existing airline reservations.
4	SR, C	Airport kiosks shall not support a Braille interface.
5	SR, NF	Check-in kiosks shall be available at airports 24x7.
6	SR, B	The System shall assign the traveler 100 FrequentTraveler credits when check-in is performed at an airport kiosk.
7	SR, FR, C	The System shall validate the FrequentTravelerID according to the following criteria: 10 characters, can be blank, alphanumeric, starts with a letter, upper-case letters, will not include the letter O.
8	SR, FR, D	The System shall use the traveler's FrequentTravelerID as the primary key for all kiosk-initiated transactions.
9	UR, NF, C	Traveler may check-in at a kiosk when flight departure time is greater than 60 minutes and less than 24 hours.
10	BR, A	Traveler check-in shall be available at all major airports by the end of next fiscal year.

Chapter Summary

In this chapter, we have developed an understanding of requirements including:

- Moving from deliverables to requirements
- The Requirements Gathering Cycle
- Defining the word “requirement”
- Categorizing requirements
- Identifying types of requirements